Audio/speech/voice applications

A lot of the time a computer is offloaded to a small machine learning accelerator to have high energy efficiency. There is often a tighter energy bound than the camera applications as well.

the foreground and background.

Semantic segmentation. This algorithm determines different parts in the image itself, such as skin, hair, trees, grass, and other aspects of a picture. This allows us to figure out the part of the image that is of interest. It is important to note that we are doing this classification at every pixel.

Content aware image enhancement. Once you have the mentioned semantic segmentation map for an image, you can enhance the colour of things easily.

Shadow removal. Often you are in a backlit situation or where you are wearing a hat that casts a shadow on your face. This can be removed by simply applying some deep learning based approaches. It is also important to keep in mind that we are trying to change the pixel value at every pixel, which requires a lot of compute.

Dehazing. Sometimes we have images that are quite hazy and foggy. To tackle that we can use some of the new state-of-the-art techniques where the conventional approaches do not work very well. If you look at the state of the art, you will see that a lot of the time it is dominated by deep learning based approaches only.

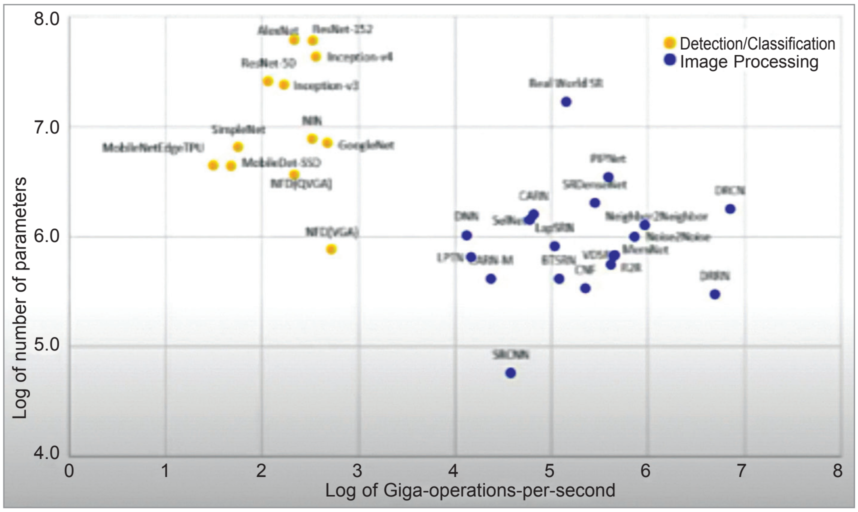

Fig. 3: Detection/classification vs image (pixel) processing

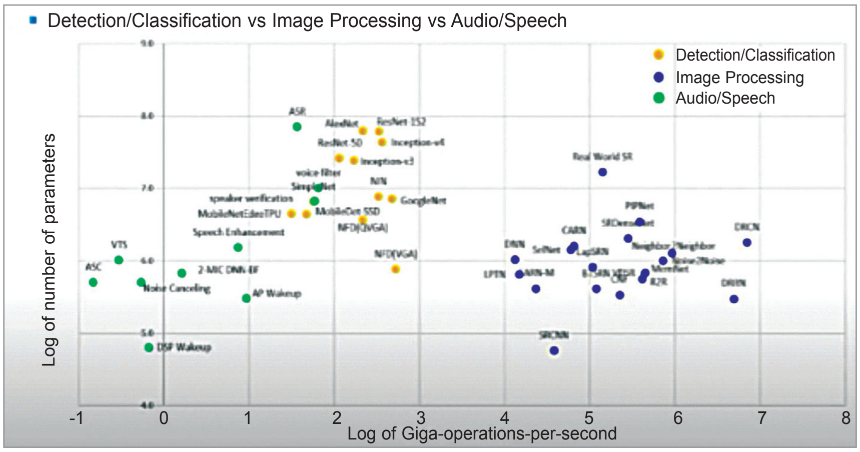

Fig. 4: Comparison of compute requirement

Comparison of compute requirements

If you look at the detection/classification that are denoted by the yellow dots in Fig. 3, the x axis is the log of giga operations per second. The blue dots are for image processing where you would either need to change the pixel values or you would need to do some classification at every pixel.

The amount of compute that you would need for those image processing techniques can be seen. The blue dots cluster and yellow dots cluster are clearly well separated. There is a huge gap between the detection/classification versus image processing.

Comparison of compute requirements

Let us take a look at Fig. 4 where the green dots are a reference for speech and audio. In all the deep learning based approaches you can find a huge gap in the compute requirement. There is about six to seven orders of magnitude difference between the green dots and the blue dots.

To summarise, it is important to understand that many multimedia applications require a lot of compute and data movement for which classical approaches are often used for higher efficiency in specialised engines. If a problem has a closed form solution then you should use the classical configurable approach. Ill posed problems may deploy DL based algorithms for high quality. Deep learning based approaches vary in compute complexity by five to seven orders of magnitude and fine-grained domain-specific processors are used for diverse workloads. EFY

This article is based on a tech talk at VLSI 2022 by Suk Hwan Lim, Executive Vice President, Head of America S.LSI R&D Center, Samsung. It has been transcribed and curated by Laveesh Kocher, a tech enthusiast at EFY with a knack for open source exploration and research.